

```
!pip install pytesseract opencv-python
!sudo apt install tesseract-ocr
```

```
Collecting pytesseract
  Downloading pytesseract-0.3.13-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (4.13.0.92)
Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from pytesseract) (26.2)
Requirement already satisfied: Pillow>=8.0.0 in /usr/local/lib/python3.12/dist-packages (from pytesseract) (11.3.0)
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)
Downloading pytesseract-0.3.13-py3-none-any.whl (14 kB)
Installing collected packages: pytesseract
Successfully installed pytesseract-0.3.13
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tesseract-ocr is already the newest version (4.1.1-2.1build1).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
```

◆ Gemini

```
import cv2
import pytesseract
from matplotlib import pyplot as plt

# Load the image
image_path = '/content/number_plate.jpg'
image = cv2.imread(image_path)

# Check if image was loaded successfully
if image is None:
    print(f"Error: Could not load image from {image_path}")
else:
    # Convert the image to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    # Apply Gaussian blur to remove noise (optional, but often helps)
    blurred = cv2.GaussianBlur(gray, (5, 5), 0)
    blurred = cv2.GaussianBlur(gray, (3, 3), 0) # Slightly reduced blur for potentially sharper edges

    # Apply thresholding to get a binary image
    # Adjust the threshold value as needed for different images
    _, thresh = cv2.threshold(blurred, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)

    # Use Tesseract to do OCR on the preprocessed image
    text = pytesseract.image_to_string(thresh, config='--psm 8') # psm 8 for single word/line
    # Define Tesseract configuration with PSM 7 and a character whitelist
    # PSM 7: Treat the image as a single text line.
    # tesseract_char_whitelist: Restrict Tesseract to only recognize uppercase letters and digits.
    tess_config = r'--psm 7 -c tesseract_char_whitelist=ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789'

    # Use Tesseract to do OCR on the preprocessed image with improved configuration
    text = pytesseract.image_to_string(thresh, config=tess_config)

    print(f"Detected Number Plate Text: {text.strip()}")
    print(f"Detected Number Plate Text (Improved): {text.strip()}")

    # Display the original image and the preprocessed image
    plt.figure(figsize=(10, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
    plt.title('Original Image')
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.imshow(thresh, cmap='gray')
    plt.title('Preprocessed Image (Thresholded)')
    plt.axis('off')

    plt.tight_layout()
    plt.show()
```

Detected Number Plate Text (Improved): TN 10AB 12

Original Image



Preprocessed Image (Thresholded)

